

Fibonacci Using Recursion

C Program

```
#include <stdio.h>

int fibonacci(int n)
{
    if(n == 0)
        return 0;

    if(n == 1)
        return 1;

    return fibonacci(n-1) + fibonacci(n-2);
}

int main()
{
    int n;

    printf("Enter number of terms: ");
    scanf("%d",&n);

    for(int i=0;i<n;i++)
    {
        printf("%d ", fibonacci(i));
    }
}
```

```

    return 0;
}

```

Input

Enter number of terms: 7

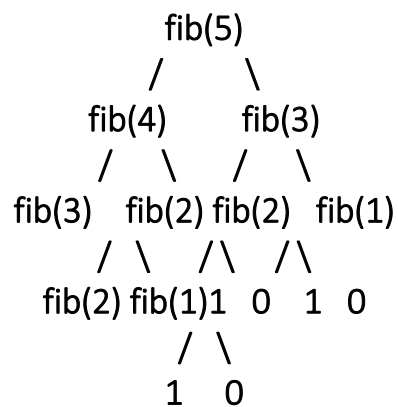
Output

0 1 1 2 3 5 8

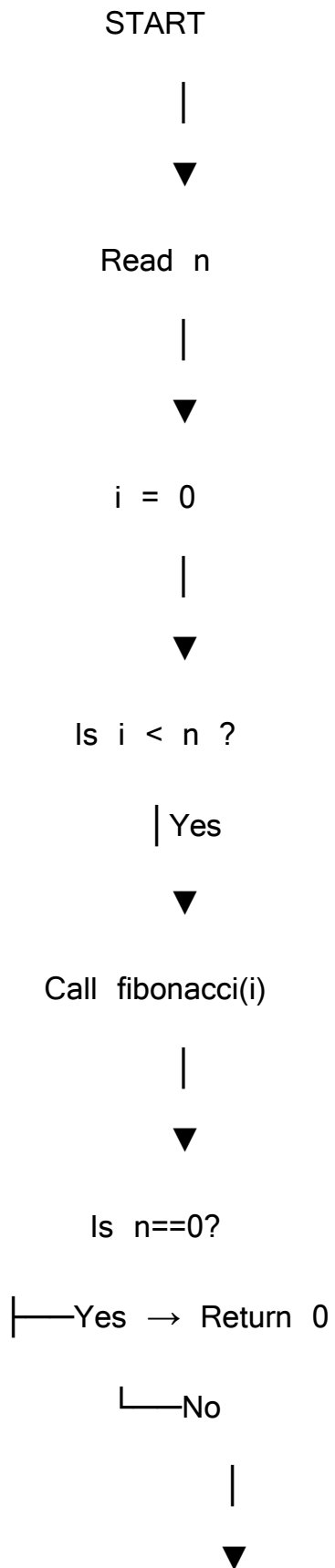
Algorithm

1. Start
2. Read n.
3. For each value from 0 to n-1:
 - Call fibonacci(i).
4. In fibonacci(n):
 - If n == 0, return 0.
 - If n == 1, return 1.
 - Otherwise return fibonacci(n-1) + fibonacci(n-2).
5. Print each returned value.
6. Stop.

Recursion Call Tree



Combined Working



Is $n=1$?

└─Yes → Return 1

└─No

|



Return fibonacci($n-1$)+fibonacci($n-2$)

|



Print Result

|



$i = i + 1$

|



Back to $i < n$?

|

No



STOP